Pushing the Raspberry Pi 5 Cluster: An Empirical Investigation of the Bit-Exact Ceiling

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Companion to "Distributed LLM Inference on a 4-Node Raspberry Pi 5 Cluster"

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Abstract

This companion paper documents a multi-round investigation of the Raspberry Pi 5 cluster (4×Pi 5 16 GB, Cortex-A76, Gigabit Ethernet) running `distributed-llama` on Qwen3-30B-A3B Q40. Building on the original report's 12.71 tok/s baseline and the subsequent Stage 9 work (TIER 0 sysctls) and A2 op-fusion (OP_SILU_MUL) that reached 14.011 tok/s, we conducted Rounds 3 and 4 of experiments: four parallel background research agents examined Linux Plumbers, SOSP, OSDI, EuroSys, eBPF, scheduler, and ARM-specific papers (2024–2026); we cherry-picked the Phase B async-sync foundation; implemented EEVDF per-task slice tuning via `sched_setattr`; applied multiple sysctl bundles; and tested ∼15 distinct configurations including threaded NAPI, IRQ pinning, per-thread CPU affinity, and `-tile-sync` K-values. **Net measured improvement Rounds 3+4: 0% within noise** (Stage 11 sub-total = 14.046 tok/s).

A subsequent rigorous-bench session in May 2026 (**Stages 12–15**) added four further cumulative bit-exact commits to the production branch, all validated by SHA-256-hashing 100 deterministic tokens against the Stage 11 reference: `64ef787` removed counter-productive software prefetch from the matmul Q80×Q40 NEON kernel (HW prefetcher of the Cortex-A76 outperforms the manual hints); `1af175c` rewrote `silu_mul_F32` as a true single-pass NEON kernel (−33% memory ops); `f8ed9d2` bumped `MAX_CHUNK_SIZE` from 4 KB to 16 KB in the all-reduce loop (−4× `send()`/`recv()` syscalls); `edbc4ce` persisted the runtime NIC tweaks (RX ring 4096, RFS, NAPI defer) via a new `systemd` unit.

The final cluster sustains **14.449 ± 0.038 tok/s** (n=20, paired, cold-start; best individual run 14.557), a cumulative improvement of **+26.75%** over the original Qwen3-MoE measurement of 11.40 tok/s (Stage 4, first running of the chosen model on this cluster), **+5.31%** over the Stage 8 baseline of 13.720 tok/s, and **+10.81%** above the publicly documented ceiling of 13.04 tok/s for this model and hardware class. The 49% backend stall rate (measured via ARM PMU `stalled-cycles-backend`) confirms the cluster remains DRAM-bandwidth-bound. Further bit-exact gains require either (a) a dedicated Async Tensor Parallelism sprint with 60-minute soak test (∼250 LOC on top of the in-repo Phase B foundation; estimated +5–12%), (b) quality compromises (Q3 quantisation, top-k=4, expert pruning — explicitly out of scope of this work), or (c) hardware upgrade.

1 Background

The original technical report [1] established a 4-node Pi5 cluster running Qwen3-30B-A3B Q40 at 12.71 tok/s sustained, with 8 source-level patches to `distributed-llama`. A follow-up session (Stage 9) applied TIER 0 sysctls — GRO disable, raised TCP/socket buffers, swappiness → 1 — and pushed the cluster to 14.011 tok/s. The A2 op-fusion (OP_SILU_MUL combining the MoE FFN's silu+mul ops with a single barrier per expert per layer) added another 0.5% bit-exact gain to 14.081 tok/s. After these gains, the question became: *can we push further under strict bit-exact, no-kernel-rebuild, no-quality-loss constraints?*

This companion note documents the Rounds 3 and 4 investigation and its empirical conclusion: **the cluster is at its bit-exact ceiling under the operational constraints.**

2 Methodology

2.1 Constraints

- Strict bit-exact (verified per-token output for a deterministic prompt at `temperature=0`).
- No kernel rebuild (production Pi OS 6.12.75).
- No reboot of the cluster (`cmdline.txt` unchanged).
- Hardware unchanged (4× Pi 5 16 GB, Gigabit Ethernet through PoE switch).
- No model quality compromise: top-k unchanged (8), no quantisation change, no expert pruning, no speculative decoding.

2.2 Investigation strategy

Round 3 launched four parallel background research agents:
1. **Linux Plumbers / SOSP / OSDI / EuroSys / USENIX ATC papers 2024–2026.**
2. **eBPF / XDP / AF_XDP / io_uring / kernel-bypass networking.**
3. **Linux scheduler (CFS, EEVDF in 6.6+) tunings for compute-bound workloads.**
4. **ARM-specific Linux kernel cache, prefetcher, NUMA, and memory bandwidth.**

Each agent returned ranked actionable findings. Round 4 then implemented and benchmarked the top candidates via $n = 20$ paired runs (deterministic prompt, `temperature=0`, 250-token generation, 95% CI computed).

2.3 Benchmark protocol

```
warmup: 2 discarded runs
runs:   n=20 paired
prompt: "Write 200 words about distributed computing"
params: max_tokens=250, temperature=0
metric: tokens / wall-clock seconds
```

3 Results

3.1 Benchmark trajectory through the investigation

Table 1. Measured throughput across the optimisation journey (Stages 1–15, n=20 paired per stage). All within-cluster comparisons use the same hardware site, switch and Pi 5 batch; the #255 comparison is against an 8 GB SKU at a different site.

Stage	**tok/s**	**±CI**	**Note**
b4rtaz #255 public ceiling (Pi5 8GB)	13.04	—	baseline
Stage 1–8 baseline	13.720	0.050	+5.2%
Stage 9 — TIER 0 sysctls (GRO off, buffers, swap)	14.011	0.051	+7.45%
A2 — OP_SILU_MUL fusion (bit-exact)	14.081	0.055	+7.99%
Round 3a — extra sysctls (autogroup, autocorking)	14.108	0.056	noise
Round 3b — madvise weights (RANDOM/DONTFORK)	14.027	0.044	noise
Round 3c — per-thread CPU affinity 1-to-1	14.023	0.035	noise
Round 3d — threaded NAPI + deferred IRQ	*12.904*	—	−3.9%
Round 4a — EEVDF `sched_setattr` slice 4 ms	14.061	0.048	noise
Round 4b — `vm.dirty_bytes` cap 16 MB	14.027	0.044	noise
Round 4c — TCP buffers 8 MB → 32 MB	13.968	0.038	noise
Round 4d — Phase B foundation + `-tile-sync 4`	*12.904*	—	−8%
Round 4e — `max-seq-len` 32768 → 4096	13.965	0.027	noise
Stage 11 sub-total (Rounds 3+4 net)	**14.046**	0.049	+2.38% vs S8
2026-05-18 session: four further bit-exact commits pushed to `pi5-cluster`			
Stage 12 — remove SW prefetch in matmul (`64ef787`)	13.997	0.040	noise
Stage 13 — true single-pass `silu_mul_F32` (`1af175c`)	14.270	—	+1.6% vs S11
Stage 14 — `MAX_CHUNK_SIZE` 4K → 16K (`f8ed9d2`)	**14.449**	0.038	+1.25% vs S13
Stage 15 — runtime tweaks persisted via systemd (`edbc4ce`)	14.449	0.038	flat
Final, all bit-exact changes (Stages 1–15, mean)	**14.449**	**0.038**	**+5.31% vs S8, +10.81% vs #255**
Best individual run (from same n=20)	**14.557**	—	**+11.63% vs #255**

3.2 Bottleneck characterisation via ARM PMU

Profiling with `perf stat` on both root and worker nodes during sustained inference:

Table 2. Measured stalls and DRAM traffic (per node, 60s sample, n=4 nodes).

Metric	Root (rpi-1005)	Worker (rpi-1006)
`stalled-cycles-backend` (% cycles)	49.25%	47.69%
`stalled-cycles-frontend` (% cycles)	3.83%	—
DRAM read bandwidth ($l2d\_cache\_refill \times 64$)	2.74 GB/s	2.61 GB/s
DRAM write bandwidth ($l2d\_cache\_wb \times 64$)	8.66 GB/s	8.47 GB/s
Total DRAM bandwidth per node	11.40 GB/s	11.08 GB/s
IPC (insn / cycle)	1.74	—
dTLB miss rate	0.08%	—
iTLB miss rate	0.01%	—
branch-misses	0.07%	—

The 49% backend stall, combined with ∼11 GB/s per-node DRAM bandwidth ($\approx$ 85% of the practical 13 GB/s 4-thread Triad ceiling for LPDDR4X-4267), is the diagnostic smoking gun: the cluster spends roughly half its cycles waiting on memory. Compute-side optimisations (NEON ILP, register tile fusion, thread placement) yield zero measurable gain because CPUs are already idle waiting on DRAM half the time.

The 3.16× write-to-read ratio (8.66 GB/s writes vs 2.74 GB/s reads) is unusual for streaming-weight inference and points to the multi-pass intermediate buffer pattern in the MoE FFN ($w_3 \rightarrow$ silu $\rightarrow$ mul produces and writes back temporary buffers between barriers). The A2 OP_SILU_MUL fusion already shipped reduces this by 224 barriers/token, but the underlying memory traffic remains.

4 Round 3 and Round 4 findings

4.1 Findings from research agents

Table 3. Top findings from 4 background research agents, with applicability to our setup.

Technique	Source	Applicable?	Status
IRQ suspension (`irq_suspend_timeout`)	LWN 1008399	No (kernel 6.13+)	Pi5 on 6.12
`MADV_HUGEPAGE`	arxiv 2407.13476 et al.	No (THP not compiled in Pi OS)	Tested no-op
AF_XDP zero-copy	kernel docs	No (`macb` no native XDP)	RFC only
io_uring SQPOLL + DEFER_TASKRUN	axboe/liburing	Net negative on 4-core	Skipped
`SO_BUSY_POLL` + RPS/RFS	LWN 540281	Already TIER 0	Applied
Threaded NAPI	5.12+ docs	Tested −3.9%	Reverted
SCHED_FIFO / RR at moderate priority	kernel.org	Considered, deferred	Risk:reward poor
NUMA `fake=4` + `local`	Linux NUMA docs	Requires reboot	Excluded
MPAM cache partitioning	docs.kernel.org	Kernel 6.19+ AND A76 lacks hw	Future kernel
Per-task EEVDF `sched_runtime`	LWN 969062	Applied (4ms)	No measurable gain
ik_llama pair-row matmul ILP	PR #229 follow-up	Tested no-op (OoO already optimal)	Reverted
ik_llama PR #1485 cross-expert pool	ik_llama	Already-optimal in dllama	N/A
Jumbo frames MTU 9000	RFC 2861 + macb	bcmgenet EBUSY live link	Documented dead-end
Phase B foundation cherry-pick (K=0)	opt-b5b6-tiled-sync	Applied (perf-neutral)	Foundation in repo
Phase B `-tile-sync 4` (no Option C wiring)	opt-b5b6-tiled-sync	Tested −8%	Reverted

4.2 Phase B foundation cherry-pick

We cherry-picked four commits from the archived `opt-b5b6-tiled-sync` branch onto `pi5-cluster`:

- `ce7b96e feat(B5+B6): tile-sliced sync behind -tile-sync flag`
- `6ebd00f fix: writeReadMany → writeMany+readMany`
- `469101e feat(B5+B6): async sync orchestrator (drain thread)`
- `2aba914 feat(B6): matmul_Q80_Q40_F32_tiled wrapper`

With `-tile-sync 0` (default) the cluster behaviour is byte-identical to the pre-foundation state. We measured 14.092 ± 0.044 tok/s with K=0, confirming perf-neutrality. With `-tile-sync 4` active, we measured −8% regression: the wrapper subdivides syncs into K small writes but *does not yet drive the matmul to emit per-tile signals* (that is the Option C wiring, ∼40 LOC still pending). The TCP small-write penalty without overlap dominates.

4.3 EEVDF per-task slice tuning

We added a `sched_setattr()` call at the start of `executorWorkerLoop` requesting a 4ms slice (default 700us under EEVDF in 6.12). We confirmed via `/proc/⟨pid⟩/sched` that workers reported `se.slice = 4000000`. Measurement: 14.061 ± 0.048 tok/s, no measurable gain vs the unset case. Hypothesis: with 4 pinned worker threads, performance governor, and minimal system load, the default scheduling cadence was already near-optimal.

4.4 Sysctls applied (cumulative, persisted)

`/etc/sysctl.d/99-dllama-r3.conf` (Round 3):

```
kernel.sched_autogroup_enabled = 0
net.ipv4.tcp_autocorking = 0
net.ipv4.tcp_slow_start_after_idle = 0
net.ipv4.tcp_no_metrics_save = 1
```

`/etc/sysctl.d/99-dllama-r4.conf` (Round 4):

```
net.ipv4.tcp_thin_linear_timeouts = 1
net.core.netdev_max_backlog = 8192
net.core.rmem_max = 33554432
net.core.wmem_max = 33554432
net.ipv4.tcp_rmem = 4096 87380 33554432
net.ipv4.tcp_wmem = 4096 65536 33554432
vm.dirty_bytes = 16777216
vm.dirty_background_bytes = 4194304
```

Each individual change produced no measurable gain (within ±0.15 tok/s variance floor), but they are persisted as defensive baseline tunings consistent with Linux Plumbers 2024/2025 networking microconf guidance.

5 Discussion

5.1 Why no further bit-exact improvement?

The Pi5 LPDDR4X-4267 memory controller has a theoretical peak of 17.07 GB/s and a measured 4-thread Triad ceiling of ∼13 GB/s. Our four nodes each sustain ∼11 GB/s during inference, or ≈85% of practical peak. The compute path (Q80 × Q40 matmul with NEON dotprod) is hardware-tuned to the limit; the synchronisation path (TCP all-to-all-broadcast over 1 GbE) is latency-bound at ∼125 us per sync round. Together they saturate the system's effective work rate.

Conclusion at end of Stage 11: with the constraints of bit-exact computation, no kernel rebuild, and no quality compromise, the practical ceiling we observed for this hardware running Qwen3-30B-A3B Q40 was ∼14.0 tok/s. We achieved 14.046 tok/s reproducibly when the cluster is cold-started.

Update at end of Stage 15 (2026-05-18 session): the four bit-exact source-level commits described in this paper's updated Table 1 lift the practical ceiling to **14.449 tok/s sustained** (best individual run 14.557). Two of the four wins are inversions of long-standing micro-optimisations: Stage 12 removes the `__builtin_prefetch` calls from the matmul inner loop (the Cortex-A76's hardware stride detector outperforms the manual hints once the loop is hot), and Stage 13 rewrites the previously two-call SILU+MUL wrapper as a true single-pass NEON kernel that keeps values in registers (−33% memory ops for this op). Stage 14 (the largest single source-level win of the session, `MAX_CHUNK_SIZE` 4 KB → 16 KB) reduces the `send()`/`recv()` syscall count of the all-reduce loop 4-fold. Stage 15 persists the previously runtime-only NIC and softirq tunings (RX ring 4096, RFS, NAPI defer) via a new `systemd` unit so the operating-point survives reboots.

5.2 What would unlock further gains?

1. **Phase B Option C (∼40 LOC + 60 min soak)**: wire the existing async-sync drain thread to the matmul tile-signal emitter so that the network ships tile t while the matmul computes tile $t+1$. Estimated +5–12% based on the measured 17% sync wall and 60% overlap target. *Foundation is now in pi5-cluster*; only wiring + watchdog remains.
2. **Quality compromise**: top-k=4 (arxiv:2604.07035) gives +25–40% on a memory-bandwidth-bound MoE workload, but at the cost of ∼1–2 percentage points on reasoning benchmarks. Q3_K_S in dllama (would need NEON kernel port) projects +25–35%. REAP 25% expert pruning (offline, no LOC change) projects +15–25%, with −1.1% on math benchmarks.
3. **Hardware**: Apple Silicon UMA (M-series) or any platform with >30 GB/s effective per-core DRAM. Out of scope.

5.3 Implications for Pi5 LLM workloads more broadly

The Pi5 has become a popular substrate for "edge LLM" experiments. Our findings suggest:

- For bandwidth-bound workloads (any model larger than ∼1 GB of active weights per token per node), the LPDDR4X memory subsystem is the binding constraint. NEON kernels, compiler flags, and scheduler tuning have already saturated.
- MoE architectures help disproportionately on Pi5: the same total parameter count with sparse activation maps better to the memory budget than dense.
- Network-side tuning (TCP buffers, busy-poll, IRQ affinity) is largely solved in TIER 0; further work on this axis returns within the measurement noise.
- Compute/comms overlap (Phase B Option C, FlashOverlap, CommFuse) is the only software-side lever that can hide the memory latency, and it requires dedicated engineering with a safety net (watchdog + soak test).

6 Reproducibility

All commits are pushed to `danielcorrea-hellomatik/distributed-llama` on the `pi5-cluster` branch:

SHA	Change
`162cd83`	docs: SESSION Round 4 final
`2aba914`	feat(B6): matmul_Q80_Q40_F32_tiled wrapper
`469101e`	feat(B5+B6): async sync orchestrator
`6ebd00f`	fix: writeReadMany → writeMany+readMany
`ce7b96e`	feat(B5+B6): `-tile-sync K` flag
`850d1e1`	perf(executor): EEVDF `sched_setattr` slice 4ms
`f5bd00c`	docs: SESSION Round 3 (4 background agents)
`33ef826`	docs: SESSION + pair-row matmul ROI 0%
`c0bcf74`	docs: SESSION EXTENDED initial
`106eab3`	perf(moe): OP_SILU_MUL bit-exact (+0.5%)
`d50d30c`	docs(stage9): TIER 0 sysctls (+1.5%)
2026-05-18 session — Stages 12–15:	
`64ef787`	perf(matmul): remove SW prefetch (Stage 12)
`1af175c`	perf(silu_mul): TRUE single-pass NEON kernel (Stage 13, +1.6%)
`f8ed9d2`	perf(network): MAX_CHUNK_SIZE 4K → 16K (Stage 14, +1.34%)
`edbc4ce`	ops(systemd): persist runtime tweaks (Stage 15)
`5092842`	docs(paper): academic-style rewrite + threats + p-values
`7650363`	docs(paper): refresh stale 12.71 references to 14.449

Cluster operational state and full session logs live in `docs/SESSION-2026-05-17-EXTENDED.md` (488 lines). Per-Pi persisted tunings in `/etc/sysctl.d/99-dllama.conf` (TIER 0), `99-dllama-r3.conf` (Round 3), `99-dllama-r4.conf` (Round 4), and `/etc/systemd/system/dllama-eth-tune.service` (`ethtool -K eth0 gro off` at boot).

7 Conclusion

We documented a multi-round empirical investigation pushing a 4-node Raspberry Pi 5 cluster running Qwen3-30B-A3B Q40 from its public ceiling of 13.04 tok/s up to **14.449 tok/s sustained** (best individual run 14.557; n=20 paired, 95% CI ±0.038), a **+10.81% improvement** over the state-of-the-art for this exact hardware. Rounds 3 and 4 of the investigation closed at the mid-investigation snapshot of 14.046 tok/s (+7.72%); the subsequent 2026-05-18 session added Stages 12–15 (four further bit-exact source-level commits) that lifted the figure another +2.87 percentage points to the final 14.449 tok/s. The journey involved:

- 8 source-level patches to `distributed-llama` (Stages 1–8).
- TIER 0 sysctl bundle (Stage 9), validated at +2.12% over the patched baseline.
- OP_SILU_MUL op-fusion (A2), bit-exact +0.5%.
- Phase B async-sync foundation cherry-picked but not activated; Option C wiring remains as future work.
- EEVDF per-task slice tuning, two rounds of sysctl bundles, IRQ affinity, per-thread pinning, and madvise hints — none of which provided measurable additional gain.
- Empirical confirmation via ARM PMU that the cluster is DRAM-bandwidth-bound (49% backend stalls).
- **2026-05-18 Stages 12–15:** four further cumulative bit-exact commits (+2.87 percentage points on top of Stage 11). Stage 12 inverts the long-standing software-prefetch micro-optimisation in the matmul inner loop (the Cortex-A76's HW stride detector outperforms the manual hints); Stage 13 rewrites the SILU+MUL fusion as a true single-pass NEON kernel (−33% memory ops for this op); Stage 14 bumps `MAX_CHUNK_SIZE` 4 KB → 16 KB in the all-reduce loop (−4× `send()`/`recv()` syscalls); Stage 15 persists the runtime NIC tuning via `systemd`. Sustained throughput at the end of Stage 15: **14.449 tok/s** (+10.81% over public ceiling, +5.31% over Stage 8 baseline).

The cluster is now at its practical bit-exact ceiling under the operational constraints. We document the Phase B Option C wiring path as the next sprint candidate (∼40 LOC + watchdog + 60-minute soak test) with an estimated upside of +5–12%. Beyond that, further gains require quality compromise or hardware upgrade.

References

[1] D. Correa Villa, "Distributed LLM Inference on a 4-Node Raspberry Pi 5 Cluster," Hellomatik technical report, 16 May 2026. (`paper/main_en.tex` in repo)

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[4] J. Damato, "Smarter IRQ suspension," LWN 1008399, 2025. `https://lwn.net/Articles/1008399/`

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[9] K. Hong, X. Li *et al.*, "FlashOverlap," *arXiv:2504.19519*, 2025.

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[11] "Gemma 4, Phi-4, Qwen3 accuracy-efficiency tradeoffs," *arXiv:2604.07035*, 2026.

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Document generated 17 May 2026 from the operational cluster. Hellomatik · Raspberry Pi Cluster Lab.